

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations
 January 2023

19IS8X38 – INTRODUCTION TO PYTHON PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Justify why Python is famous programming language by explaining any five features of python.	10	L*3	1	1
	b) What are Lists? How are they created and accessed?	6	L1	2	2
	c) Write a Python Program to calculate Simple Interest.	4	L2	1	2
2.	a) Illustrate Any 5 string operations strings in python with example.	10	L2	1	1
	b) Write a program to count the number of occurrence of each character in a string and number of occurrence of words and number of vowels in a sentence.	10	L2	1	2
3.	a) Explain any 5 Dictionary methods in python.	10	L4	2	1
	b) Explain any five types of list operations with example.	10	L2	2	2
Unit – II					
4.	a) Define constructor, its types with general syntax and types of constructors with suitable code.	10	L2	3	1
	b) Write a binary search function using iterative and recursive methods which searches an item in a sorted list. The function should return the index of the element to be searched in the list.	10	L2	2	2
5.	a) What is inheritance? What are its features? Explain. Demonstrate how inheritance is done in python with an example program.	10	L2	3	1
	b) Describe Functions with Default values and Default arguments in Python.	10	L2	2	2
6.	a) Does python support operator Overloading? What is its advantage, write a code to overload greater than operator.	8	L3	3	3
	b) Write a python program to find the area of a rectangle using classes, create suitable methods, create objects to demonstrate the working of class methods.	6	L2	2	3
	c) What are exceptions? Discuss the different techniques used to handle the exceptions in python.	6	L2	3	2
Unit – III					
7.	a) Discuss any five widgets of tkinter widgets with examples.	10	L2	4	2
	b) Explain the following methods with respect to file handling in Python with suitable sample code. i. seek() ii. readline()	10	L2	5	2
8.	a) Design a suitable python GUI using tkinter to make use of Label, button widgets.	10	L3	4	3
	b) Write python programs for the following. i. Count Number of Files in a directory ii. Delete specific lines from file	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
